

PassGig

Economic Passport + Skill-to-Gig Platform

Squad Hackathon 3.0 — Challenge 02

System Design Brief & Team Reference Document

TARGET USERS	CORE TECH	SQUAD API ROLE	CHALLENGE
Unemployed Youth + Informal Traders	AI Matching & Credit Scoring	Payments, Escrow & Transaction Data	Challenge 02 Smart Systems

Prepared for internal team alignment · PassGig · 2025

What is PassGig?

PassGig is an intelligent economic platform that connects unemployed Nigerian youth with gig opportunities posted by informal traders and artisans — while simultaneously building a financial identity for both sides that unlocks access to credit, savings, and insurance. It is built specifically for Nigeria's informal economy.

The Core Problem We Are Solving

Nigeria has over 13 million unemployed youth and a sprawling informal economy of traders, artisans, mechanics, tailors, food vendors, and repair workers. These two groups exist in the same neighborhoods but are disconnected. Traders cannot find reliable help. Youth cannot prove their skills or get paid reliably. Neither group has the financial history needed to access formal credit or insurance — because the formal system was never built for them.

PassGig fixes the connection layer and, in doing so, generates the transaction and behavioral data that builds financial identities from scratch.

Two User Types. One Ecosystem.

USER TYPE A	USER TYPE B
Unemployed Youth	Informal Traders & Artisans
• Age 18–30, smartphone user • Has skills but no proof of them • Has no employment history or payslip • Active on WhatsApp, Instagram, TikTok • Wants income, experience, and recognition • Goal: Get gigs, build reputation, unlock financial access	• Shop owners, tailors, mechanics, food vendors, electricians, carpenters, artisans • Has a business but runs it entirely informally • Needs reliable task workers but has no hiring system • Has consistent cash flow but no financial profile • Goal: Find help, pay reliably, access business credit

Key insight: The informal trader is BOTH a customer and a data source. Every time they hire a youth worker and pay through PassGig, they generate transaction data that builds their business credit profile — without them doing anything extra.

The Three Layers of PassGig

The platform is built in three interconnected layers. Each layer feeds the next, creating a compounding system that gets more valuable as more people use it.

Layer 1 — Onboarding & Economic Passport

Every user — youth or trader — creates an Economic Passport when they join PassGig. This is a dynamic digital profile that starts with declared information and grows richer over time through real activity on the platform.

For Youth:

- Full name, age, location (LGA level)
- Skills declared: e.g. Graphic Design, Plumbing, Delivery, Tailoring, Carpentry
- Availability: days/hours per week
- Portfolio upload: photos, certificates, references (optional at start)
- Initial trust score generated by AI from declared data and phone verification

For Traders / Artisans:

- Business name, trade type, location
- Nature of tasks they typically need help with
- How often they hire, rough budget range
- Initial trust score from business verification signals

The passport is not static. Every completed gig, every payment made on time, every positive rating received — all of it updates the passport score in real time. The passport becomes a living financial identity, not just a one-time registration.

Layer 2 — The Gig Matching Engine

Traders post gigs on the platform. The AI matching engine then finds the best-fit youth workers based on multiple signals and ranks them for the trader to review and hire.

Matching Signal	What It Does
Skill alignment	Compares the gig requirements to youth declared skills
Location proximity	Prioritizes youth within the same LGA or nearby area
Passport score	Ranks higher-trust workers above new or low-rated ones
Availability match	Only shows workers free during the gig's required hours

Past gig ratings	Weights workers who have positive reviews from prior gigs
Gig category history	Favors workers with experience in that specific trade

Once a match is confirmed, the trader deposits payment into escrow via Squad API. The youth worker is notified, accepts the gig, completes the work, and the trader releases payment. Both sides rate each other. Both passports update.

Layer 3 — Intelligence & Financial Inclusion Loop

This is where the platform becomes more than a job board. The data generated by completed gigs and Squad transactions feeds an intelligence layer that unlocks real financial services for both user types.

User	What They Unlock	How They Unlock It
Youth Worker	Micro-savings account Small personal credit line Skill certification badge	Complete 5+ gigs with 4+ star ratings Consistent Squad payment history Pass skill assessment quiz
Trader / Artisan	Working capital loan Business insurance referral Verified Trader badge	3+ months of hiring activity Consistent payment behavior Good worker ratings given

How Squad Powers the Core of PassGig

Squad API is not a token integration added at the end. It is structurally embedded into three critical functions of the platform:

Squad API Function	How PassGig Uses It	Why It Matters
Payment Escrow	Trader deposits gig payment upfront into escrow	Prevents both parties from walking away without payment
Worker Disbursement	On gig completion and trader confirmation, Squad releases funds directly to worker's account	Fast, reliable payment to workers
Transaction Data Feed	All payments flow through Squad, generating a structured transaction history per trader	This structured transaction history is the data that powers the AI
Business Payments	Traders can use Squad to receive payments from their broader customer base, signed to their PassGig profile	Adds a broader revenue stream to the trader

Disqualification note from the brief: A solution with no meaningful Squad API integration will not be eligible for top placements regardless of scores in other criteria. PassGig's escrow and disbursement architecture ensures Squad is central — not cosmetic.

What the AI Actually Does

The AI layer has three distinct, concrete jobs in PassGig. Each one is technically demonstrable at the hackathon:

1. Trust Scoring Engine

Generates an initial passport score at onboarding using declared data, phone verification, BVN signals (optional), and device signals. Updates the score dynamically after every completed gig, payment event, and rating. This is a classification/regression model trained on behavioral and transactional signals — similar to alternative credit scoring used by fintechs like Fairmoney and Carbon, adapted for the informal economy context.

2. Gig Matching Algorithm

Ranks available workers for each posted gig using a weighted scoring function across skill fit, location, passport score, availability, and category experience. Can be implemented as a collaborative filtering model or a simpler weighted ranking system depending on data volume at demo stage. Outputs a ranked shortlist for the trader.

3. Skill Gap & Demand Recommender

Analyzes gig posting patterns across locations and categories to identify unmet demand. For example: if there are 40 gig postings for electricians in Surulere but only 5 electricians registered, the system flags this and recommends nearby youth workers to consider upskilling. This is the system 'learning and improving over time' the challenge brief specifically asks for.

06 END-TO-END USER FLOW

How a Gig Works — Step by Step

#	Actor	Action	System Response
1	Youth	Downloads PassGig app, signs up, declares skills and location	AI generates initial passport score. Profile created.
2	Trader	Signs up, describes business, posts a gig (e.g. 'Need delivery driver Saturday, 13,000')	Gig published. Trader activates.
3	System	Matching engine runs	Top 5 youth workers ranked and shown to trader based on skill, location
4	Trader	Reviews shortlist, selects a worker, confirms the gig	Worker notified. Trader prompted to deposit 13,000 into Squad escrow
5	Trader	Pays into escrow via Squad API	Funds held securely. Worker sees 'Payment secured — gig confirmed'
6	Youth	Shows up, completes delivery	Trader marks gig complete on app.
7	System	Squad API releases funds to worker wallet/account	Worker receives payment instantly. Both parties prompted to rate each other
8	Both	Submit ratings	AI updates both passports. Scores adjust. Transaction logged.
9	System	Intelligence layer analyzes data	Demand signals updated. Credit eligibility recalculated. Skill recommendations

07 SUGGESTED TECH STACK

What to Build With

Layer	Technology / Tool	Purpose
Frontend (Web)	React.js + TailwindCSS	Main app interface for both user types
Frontend (Mobile)	React Native or Flutter	Mobile app — primary access point for youth
Backend	Node.js (Express) or Python (FastAPI)	API server, business logic, Squad integration
Database	PostgreSQL	User profiles, gig records, transaction logs
AI / ML	Python (scikit-learn / TensorFlow Lite)	Trust scoring model, gig matching algorithm
Squad API	Squad Payment APIs	Escrow, disbursement, transaction data
Auth	Firebase Auth or JWT	User authentication and session management
Hosting (Demo)	Render / Railway / Vercel	Fast free-tier deployment for hackathon demo

Mapping to the Challenge 02 Rubric

Criteria	Weight	How PassGig Addresses It
Squad API Integration	25%	Escrow, disbursement, and transaction data are all Squad-powered. Squad is in the c
Technical Architecture	20%	Three-layer architecture: passport onboarding → AI matching → financial inclusion. C
Problem Understanding & Innovation	20%	Addresses both sides of the informal economy problem. The feedback loop between g
Economic Viability & Scalability	20%	Commission per gig (5–8%) is the revenue model. Scales nationally by adding city hu
Presentation & Communication	10%	Two distinct user journeys make the demo concrete and relatable. Both can be shown
Impact Potential (Bonus)	10%	Targets 13M+ unemployed youth. Informal economy is 65% of Nigeria's GDP. Both g

09 RECOMMENDED NEXT STEPS FOR THE TEAM

What to Do Now

1. Align on scope for the demo

Decide: are you showing both the youth flow AND the trader flow, or focusing on one? Both is ideal for impact, but one working end-to-end is better than two broken ones.

2. Set up Squad API sandbox

Get API keys and test the escrow and payment flow immediately. This takes the most integration time and is the highest-weighted judging criteria.

3. Build the passport & matching MVP first

Onboarding form → AI score → gig posting → shortlist display → escrow payment. That is your demo spine.

4. Assign the AI model early

Even a simple weighted scoring function qualifies. You need someone on the team focused purely on the trust scoring and matching logic from day one.

5. Talk to at least 3 real users

The rubric explicitly asks: have you spoken to your target users? One unemployed youth, one market trader, one artisan. Even WhatsApp voice notes count as evidence.

6. Prepare the one-pager

The judges keep this after your presentation. It should be: problem, solution, Squad APIs used, four pillars addressed, impact numbers. One A4 page. Design it properly.

Remember: A working demo beats a perfect slide deck. Build the core loop — user signs up → gig posted → match made → Squad payment flows → passports update — and make sure it runs live without breaking. That is your winning condition.